INTERNATIONAL UNIVERSITY OF APPLIED SCIENCES

DLMDSEBA01 Business Intelligence I

CASE STUDY

Meier Drogeriemarkt GmbH

Author: Tardzenyuy Emmanuel Yufenyuy

Enrollment Number: IU14091118

Study Program: Business Intelligence

Contents

Abstract 1
Introduction 1
Key Terminologies 1
Selecting the right Extract-Transform-Load (ETL) Tool 3
Selected Criteria 4
Selected ETL Tools for Comparison 5
Talend Open Studio 5
Pentaho Data Integration / KETTLE 5
Microsoft SQL Server Integration Services (SSIS) 6
Apata OS Data Integration 6
ETL Integrator 6
Choosing the right Business Intelligence (BI) Tool 6
Description of Business Intelligence Tools 7
Power BI 7
Tableau 7
Quick View 8
IBM Cognos Analytics 8
SAS Visual Analytics 8
Suggested Data Warehouse Architecture for Meier Drogeriemarkt GmbH 9
ETL Process at Meier Drogeriemarkt GmbH Using Talend 9
Extraction 10
Transformation 10
Loading 10
Modelling a Datamart at Meier Drogeriemarkt GmbH 10
Reporting with Tableau 11
Challenges 12
Conclusion 12

List Of Figures

Fig._ 1: ETL-Software Selection Criteria (Hanine et al. SpringerPlus 2016) 3
Fig._ 2: Referenced architecture (Dr. Vaas, Silke .Business Intelligence I[Course Book].IU International University of Applied Science. https://www.iu.org/ 2024). 9
Fig._ 3: Star Schema Architecture for Sales and Marketing Departmental Data Mart at Meier Drogeriemarkt GmbH. 11

Abstract

Meier Drogeriemarkt GmbH, a fast-growing drugstore based in Cologne, Germany, is embarking on a business intelligence(BI) project to streamline its strategic decision-making process through data-driven insights. This initiative involves the development of a unified reporting system built on a data warehouse architecture that consolidates data from multiple sources into a central repository to enhance the calculation of the company's key figures that will serve as a single source of truth for every stakeholder with a decisive role. The architecture incorporates essential components such as source systems, data marts, an Extract-Transform-Load (ETL) process, and analytics tools to meet the requirements set by management. A reference architecture is proposed to guide the creation of a tailored data warehouse solution for Meier Drogeriemarkt GmbH. The establishment of this unified system will provide data resources to branch managers to identify those customers and product range that create the most value for Drogeriemarkt. This report addresses the selection of appropriate ETL and business intelligence tools, along with the design and presentation of the data warehouse architecture and its components, to support sustainable and efficient data processing and analysis.

Introduction

Meier Drogeriemarkt GmbH is a prominent drugstore located in western Germany, with its headquarters in Cologne and several branches across the country. With the growing volume of data being generated, the company recognizes the need to enhance its strategic decision-making process to achieve business objectives. In response, management has initiated a business intelligence project focused on developing a unified reporting system. This system aims to centralize data into a single repository, facilitating the analysis of key performance indicators. The foundation of this system lies in a referenced data warehouse architecture comprising several core elements, including source systems, a data warehouse, data marts, an ETL process, and analytics tools. By leveraging frameworks as references, the proposed architecture will align with the business goals established by management. To ensure the architecture's sustainability, the development process emphasizes the selection of efficient ETL and business intelligence tools. The report is structured to provide a comprehensive approach, starting with defining a few key terminologies, outlining business requirements, comparing and choosing the suitable ETL and BI tools. This will be followed by a detailed design of the data warehouse architecture that fits the context of Meier Drogeriemarkt GmbH, explanation of the ETL process, a logical representation of a data mart in a star schema architecture, and how the chosen BI tool will be used for analytics. Finally, challenges faced in the course of this work and concluding statements will summarize the proposed solution for this task.

Key Terminologies

- Data warehouse(DWH): This is a database implementation that supports the collection and analysis of historical data (Biplob, M. B., Sheraji, G. A., & Khan, S. I. ,2018). Data in a data warehouse is persistently stored by theme and this is non-volatile. Data warehouses are used in Online Analytical Processes due to performance gain via response time as data in data warehouses are stored in de-normalized form(Sreemathy, J., Nisha, S., & RM, G. P., 2020, March). Other features of a data house include connection to data sources, data collection, administration, analysis, and presentation.
- Data mart: It is also known as an evaluation database and contains data for a particular purpose typically for a specific user group or department.
- Staging Area: This is a temporal store that serves as a transition area between the data sources and the data warehouse. During the transition, data from the source is transformed before it is moved to the data warehouse.
- Data Integration: It is the process of transforming data of different types into a unified format.
- Data ingestion: It is the action that writes data into a schema of a database or data warehouse. This includes update and insert operations.
- Extract – Transform – Load: It is a process of Extracting data from different source systems and transforming the data into a unified agreed format then loading the transformed data into the data warehouse using different tools (Sreemathy, J., Nisha, S., & RM, G. P., 2020, March). Extraction can either be fully, partially with or without update notifications. The loading process to refresh

the data can either be incrementally, fully, or by initial load (Sreemathy, J., Nisha, S., & RM, G. P., 2020, March).

- Star Schema: It is an architecture that is used in the physical implementation of a data marts in data warehouses. In this schema, there is a central table called the fact table. This is surrounded by other tables called dimension tables. The relationships between the fact table and its dimension tables are achieved by inserting the primary keys of the dimension tables into the fact table as foreign keys.
- Facts: These are concrete numerical values often obtained as results of measurable(s). The values are meaningless or cannot be interpreted without features from the dimension tables. An example of a fact is 13.091.
- Key figures: These are measures of features that are calculated to yield facts. An example of a key figure is Sales and a measure of sales can be Average sales.
- Branch management: This is a major area of concern for most financial institutions and businesses because failures at the level of branches can have a significant negative impact on marketing and subsequently affect the survival of the business(Deng, S., Moutinho, L., & Meidan, A. 1991). Studies shows that branch managers can improve the performance of branches by engaging in business activities that include, the evaluation of branch weaknesses and strengths, analyse the ability and competence of branch Personnels in effectively serving branch customers, analyse the physical impact and location of the branch etc(Deng, S., Moutinho, L., & Meidan, A. 1991). Additionally, a central function of branch managers is customer retention. This can be achieved by involving branch managers in the development of marketing strategies that will provision customers with suitable products and services (Deng, S., Moutinho, L., & Meidan, A. 1991).
- Category Management: Among other definitions, “Category management is a process that involves managing product categories as business units and customizing them on a store-by-store basis to satisfy customer needs” (Nielsen, 1992, as cited in Dussart, C.,1998). It is an intensive business process that is data-driven, and fact-based and involves the analysis of full streams of data from different sources having different characteristics. In category management, individual items are grouped into product categories instead of manufacturer's brands. This is alignment with the tendency of consumers to be more likely to purchase within the same product category (Dussart, C.,1998).
- Customer Relationship Management: A process of creating and maintaining relationships with customers that will enable businesses to identify, attract, differentiate and retain customers(Hassan, R. S., Nawaz, A., Lashari, M. N., & Zafar, F. 2015). Through CRM businesses can effectively coordinate departments like sales, customer service, marketing, field support and other departments that have touch points with customers. CRM creates sustainable competitive advantage for businesses as they can curate their services and product offerings to suit customer needs (Hassan, R. S., Nawaz, A., Lashari, M. N., & Zafar, F. 2015).

Selecting the right Extract-Transform-Load (ETL) Tool

The selection of the right ETL tool is a vital but complex event in the designing and implementation of a business intelligence project. The complexity is a result of the factors that need to be considered which include the business requirements that must be fulfilled by the tool(s) as well as the numerous ETL tools that are available to choose from. ETL tools can be categorized into open-source ETL tools and commercial (Szilveszter, M., Molnár, G., & Nagy, E., 2024, April). Open-source ETL tools are available for free under different licenses depending on the level of use and distribution (Thomsen, C., & Pedersen, T. B. , 2005). In the need of more resources, open-source ETL tools require a commercial license which is provided by the owner of the tool. Alternatively, a commercial ETL tool can be chosen which is only available under a paid license. In practice, basic conditions considered by professionals when choosing a tool include the availability of *commercial support, training, and consulting services* (Thomsen, C., & Pedersen, T. B. , 2005).

Hanine *et al. SpringerPlus, (2016)* suggests a multi-criteria-decision method based on AHP(Analytical Hierarchy Process) and TOPSIS (Technique for Order Preference by Similarity to Ideal Solution). Fig._1 shows the main and sub-criteria used for ETL tools comparison during the process of finding the best ETL tool for a business intelligence project. It is recommended that companies should consider the flexibility of the ETL tool as this is important for adaptability in case of changing business conditions.

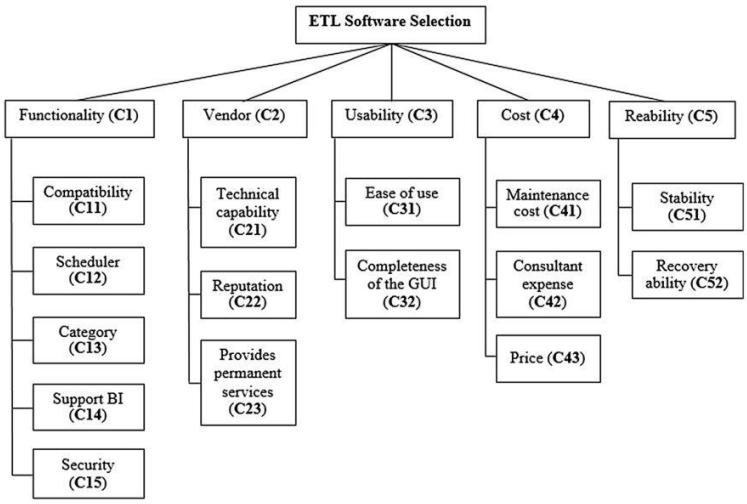

Fig._ 1: ETL-Software Selection Criteria (Hanine et al. SpringerPlus 2016)

According to Biplob, M. B., Sheraji, G. A., & Khan, S. I. (2018, October) the criteria to be used for comparing ETL tools include: Ease of use, Performance, Functionality, Reusability, Usability, Automatic recovery of flows or workflows, multi-role team collaboration, clustering and job distribution, large volume performance, connectors, and data partitioning. These criteria were used to compare the following the ETL tool: Pentaho kettle enterprise, Talend integration suit, Informatica power center, pervasive data indicator, and Microsoft SQL server integration services.

From the perspective of Szilveszter, M., Molnár, G., & Nagy, E. (2024, April), the most prevalent ETL tools are Microsoft SQL server integration services, Talend open studio, and Pentaho Data integration. For comparing these tools, the following criteria were used: functionality, performance, scalability, User-friendliness, extensibility, connectors, plugins, and contribution from the community are the criteria that qualify to be used in the comparison of ETL tools.

Whereas Majchrzak, T. A., Jansen, T., & Kuchen, H. (2011, March) consider the following as the major criteria for ETL tool comparison: performance(the main measurements or metrics for performance are execution time, CPU load, and memory load), integration, interfaces, graphical editor(ease of use), functionality, support and documentation, up-to-datedness. These criteria were intended used to compare the following ETL tools: Apatar OS Data Integration, CloverETL Engine, Enhydra Octopos, Jitterbit Integration Environment, KETL, Pentaho Data Integration, Scriptella, and Telend Open Studio. Based on their comparison, Talend Open Studio and Pentaho data integration fulfilled all the set criteria but then further comparison of these two based on metrics for measuring performance showed that Talend Open Studio is a better choice over Pentaho data integration. This is a result that Talend Open Studio performs better in less time with comparatively lesser CPU usage than Pentaho data integration.

Considering the business requirements of Drogeriemarkt GmbH, a few criteria and ETL tools mentioned above have been chosen. This is to reduce the number of criteria and ETL tools to analyze based on the business requirements. For the comparison, the aspect of open source and commercial licensing is overlooked. The selected described criteria are below.

Selected Criteria

- Functionality: The ability of the tool to integrate third-party libraries is very important (Majchrzak, T. A., Jansen, T., & Kuchen, H., 2011). Additionally, it should be clear to the user whether the ETL tool is data purging or data change arranged or performs both (Biplob, M. B., Sheraji, G. A., & Khan, S. I., 2018). This aspect of the tool provides room for adaptability.
- Performance: This aspect measures the effectiveness of the ETL tool in implementing its tasks. For instance, the amount of time and memory usage with which a tool handles a given amount of data (Majchrzak, T. A., Jansen, T., & Kuchen, H., 2011).
- Connectors: An ETL Tool should be able to connect to several different data sources e.g SAP, Salesforce, FTP, FIX, Cloud, Sharepoint, Database, etc. (Biplob, M. B., Sheraji, G. A., & Khan, S. I., 2018). This is relevant because the core aspect of a data warehouse is to centralize data from different data sources.
- Parallel Job Distribution: This is particularly important when the ETL tool has to deal with large volumes of data(Biplob, M. B., Sheraji, G. A., & Khan, S. I., 2018). This functionality allows load distribution across several servers and is thus necessary in the data transformation.

- Ease of use: The ETL tool should have a graphical user interface that will allow users to interact with the tool and perform tasks like administrating data maps (Majchrzak, T. A., Jansen, T., & Kuchen, H., 2011).
- Support: This aspect enhances the ease of use and personal development of the users. This includes the availability of user manuals, training possibilities, developer communities, consulting services, and documentation (Biplob, M. B., Sheraji, G. A., & Khan, S. I., 2018).

Selected ETL Tools for Comparison

In data warehousing, it is very important to know whether its implementation is achieved with Relational Online Analytical Processing (ROLAP) or Multidimensional Online Analytical Processing (MOLAP). In ROLAP, the ETL tools load relational database tables whereas in MOLAP the tools load multidimensional cubes. In practice, it is also important to how data will be loaded into the data warehouse. Incremental loading is often preferable (Majchrzak, T. A., Jansen, T., & Kuchen, H., 2011). There is also a hybrid way of loading data into the warehouse. This is called Hybrid Online Analytical Processing (HOLAP) and it seeks to exploit the advantages of ROLAP and MOLAP. The five ETL tools selected for comparison against the criteria described above are elaborated as follows.

Talend Open Studio

Talend Open Studio offers extensive support for data integration through training, consulting, and detailed guides. It is primarily designed to support ROLAP, with the flexibility to output data to Palo MOLAP, and enables seamless data loading from diverse sources like flat files, DBMSs, web services, CRM systems, Cloud Services, and other systems. This functionality is particularly important for data extraction. Users can specify ETL jobs via its graphical user interface (GUI), which simplifies the process while generating Java and Perl code automatically for execution. Talend also includes several transformations for data quality and functionalities like matching, replacing, etc. Additionally, it supports slowly changing dimensions and bulk loading and facilitates incremental updates through lookups for efficient inserts or updates (Thomsen, C., & Pedersen, T. B., 2005; Majchrzak, T. A., Jansen, T., & Kuchen, H., 2011). In ETL Processes, Talend open studio uses less amount of time and less memory usage in processing different volumes of data. This is particularly important when performance is a considerable factor (Majchrzak, T. A., Jansen, T., & Kuchen, H., 2011).

Pentaho Data Integration / KETTLE

Pentaho Data Integration (formerly Kettle) is a powerful ETL tool acquired by Pentaho to form Pentaho Kettle Enterprise. It is available under the LGPL (Lesser General Public License), making it an accessible solution for a variety of users. The platform supports ROLAP while also enabling data movement from ROLAP to MOLAP for enhanced analysis capabilities. Data can be loaded from flat files, and a commercialized SAP connector is available for seamless integration with SAP systems. Pentaho provides comprehensive support through training, consulting, user manuals, and an active support forum. The platform offers a variety of pre-processed

transformations that can be designed via an intuitive graphical user interface (GUI). Executed transformations are logged in files with timestamps, allowing for effective tracking and facilitating incremental updates. Additionally, jobs can be executed in parallel, enhancing processing efficiency for complex workflows (Thomsen, C., & Pedersen, T. B., 2005). During an ETL Process, the amount of memory usage by Pentaho data integration is quite constant with high amounts of processing time. This applies to varying sizes of data sets being processed in ETL (Majchrzak, T. A., Jansen, T., & Kuchen, H., 2011).

Microsoft SQL Server Integration Services (SSIS)

Microsoft SQL Server Integration Services (SSIS) is a robust ETL tool embedded with several built-in transformations, enabling the extraction of data from various sources. It provides powerful, flexible, and customizable solutions for data transformation and is designed to be easy to develop and maintain. SSIS also integrates seamlessly with other products in the Microsoft stack, making it a versatile choice for organizations using Microsoft technologies. However, the learning curve for SSIS is significant, requiring considerable effort before users can fully leverage its capabilities. Additionally, debugging processes can be challenging, demanding careful attention and expertise (Majchrzak, T. A., Jansen, T., & Kuchen, H., 2011).

Apata OS Data Integration

Apata OS Data Integration is an open-source ETL tool available under a GNU and GPL license, with a commercial license option for additional services. It emphasizes ease of use by offering training, consulting, and documentation, along with a public forum where users can seek assistance. The platform exclusively supports ROLAP and allows data to be loaded from sources such as CSV files, Excel spreadsheets, ERP systems, and CRM platforms. Data updates can be managed through its graphical user interface (GUI), though it does not support incremental updates (Thomsen, C., & Pedersen, T. B., 2005).

ETL Integrator

ETL Integrator is a free tool available under the Common Development and Distribution License (CDDL). It offers comprehensive documentation, including video tutorials, to support users. The platform exclusively supports ROLAP and provides a range of pre-processed transformations and data-cleansing operators/functions. ETL Integrator enables incremental updates and supports the parallel execution of jobs for efficient processing. Data can be loaded from files and ERP systems, making it a versatile choice for data integration tasks (Thomsen, C., & Pedersen, T. B. , 2005).

Comparing the above elaborated ETL tools based on functionality, performance, connectors, parallel job distribution, ease of use, and support, and alongside the requirements from Drogeriemarkt GmbH, Talend Open Studio stands out and has been suggested as the best tool to fulfill these requirements. These tools will thus be used in the implementation of the ETL process in the data warehouse architecture for Drogeriemarkt GmbH.

Choosing the right Business Intelligence (BI) Tool

The brain of an average human being is built to quickly capture information in visual form. This is the foundation of visualization and data visualization tools are developed on this premise which enables them to convert complex and large volumes of data into visuals. Thus making it easy for an audience, irrespective of their expertise, to understand and interpret rather than barely looking at the data rows. From a user perspective, a data visualization tool stands out when it is *User-Friendly*(Skender, F., & Manevska, V. 2022). For this reason, most business intelligence tools available in the market are developed to meet up to this requirement(Skender, F., & Manevska, V. 2022; Lavanya, A., Sindhuja, S., Gaurav, L., & Ali, W. 2023). Other factors to consider when choosing a business intelligence tool include: *data complexity, desired output, and needed level of interactivity with the data*(Lavanya, A., Sindhuja, S., Gaurav, L., & Ali, W. 2023). Analysts can leverage business intelligence tools to quickly identify focus areas through exploratory data analysis and share found insights, such as trends, outliers, and patterns, with business managers to enable them to make data-driven decisions. With regards to the requirements of Meier Drogeriemarkt, it is important to choose a business intelligence tool that will get the most insight out of the enriched data stored in the data warehouse. To make a choice, it is important to have an in-depth understanding of the main functionalities of the business intelligence tools available in the market. As such a literature review of some of these tools will be done. This includes:

Description of Business Intelligence Tools

Power BI

Power BI is a versatile business intelligence tool from Microsoft's robust technology stack, designed to help businesses efficiently generate actionable insights. It supports multiple data source connectors, enabling users to extract and independently transform data from various sources. In the model view of its desktop component, relationships between data tables can be established to enable filtering across the tables. Its friendly facilitates the creation of real-time dashboards, and interactive reports through its advanced visual capabilities such as line, pie, doughnut, matrix, map, scatter, and bar charts, providing businesses with a comprehensive view of their key performance indicators (Lavanya, A., Sindhuja, S., Gaurav, L., & Ali, W. 2023). This tool also provides drill-down and roll-up possibilities to allow users to navigate the data hierarchically. Additionally, Power BI integrates seamlessly with programming languages like R, Python, and SQL, empowering advanced users to incorporate custom scripts and analytics into their projects(Skender, F., & Manevska, V. 2022). Besides the Power BI desktop, there also exist other components for cloud services like Power BI service and Power BI apps which makes it suitable for self-service and enterprise-level use. (Skender, F., & Manevska, V. 2022). With the ability of Power BI to leverage deep analysis of data to enhance data-driven decision-making, Power BI has established itself as a leading choice in the business intelligence space.

Tableau

Tableau is a great data visualization and analytics tool offering both public and commercial licenses, with the paid version providing advanced functionalities. It enables users to create different and interactive visuals with ease without requiring

programming expertise(Skender, F., & Manevska, V. 2022). Key features include drag-and-drop functionality, interactive tables, data highlighting, filtering, commenting, and customizable views. Tableau is designed to handle large datasets with high speed and capacity, supporting direct queries and streaming for real-time insights(Skender, F., & Manevska, V. 2022). Users can also edit data using grouping and clustering properties. Tableau simplifies complex data analysis, making it accessible and efficient for businesses(Skender, F., & Manevska, V. 2022; Lavanya, A., Sindhuja, S., Gaurav, L., & Ali, W. 2023).

Quick View

QlikView is a high-performance data analytics and visualization tool suitable for businesses of all sizes and industries. It is designed to be user-friendly while offering powerful capabilities, making it an ideal choice for applications in artificial intelligence and machine learning. QlikView provides diverse visualization options and can seamlessly handle large datasets, offering a robust cloud platform for scalability(Lavanya, A., Sindhuja, S., Gaurav, L., & Ali, W. 2023). With its real-time analytics capabilities, it bridges the gap between information and action, enabling businesses to respond swiftly to insights. Qlik View is suitable for artificial intelligence and machine learning(Lavanya, A., Sindhuja, S., Gaurav, L., & Ali, W. 2023). Its precise and clear documentation further simplifies usage, and it offers the ability to uncover new relationships between data points, driving deeper understanding and informed decision-making(Lavanya, A., Sindhuja, S., Gaurav, L., & Ali, W. 2023).

IBM Cognos Analytics

IBM Cognos Analytics is a self-service business intelligence and analytics tool suitable for organizations of all sizes. It offers AI assistance to enhance data analysis, enabling users to uncover patterns such as customer behaviour, market trends, and business performance across various periods, regions, and business lines. Its robust reporting feature helps organizations gain actionable insights to make data-driven decisions(Lavanya, A., Sindhuja, S., Gaurav, L., & Ali, W. 2023). However, users have occasionally raised security concerns. Despite this, IBM Cognos Analytics remains a powerful tool for discovering trends and optimizing business outcomes(Lavanya, A., Sindhuja, S., Gaurav, L., & Ali, W. 2023).

SAS Visual Analytics

SAS Visual Analytics is a scalable, easy-to-use analytics tool designed for businesses of all sizes. It offers a powerful in-memory environment, making it suitable for self-service analytics, advanced reporting, and robust data governance. As an all-inclusive platform, it helps uncover links and patterns in data, providing actionable insights. Data can be loaded directly into servers without the need for complex data pipelines, streamlining the analysis process. Ideal for industries like insurance, SAS Visual Analytics empowers organizations with advanced analytics and reporting capabilities to drive smarter decision-making(Lavanya, A., Sindhuja, S., Gaurav, L., & Ali, W. 2023).

Considering the business requirements from Meier Drogeriemarkt and user-friendliness as a key factor for users in choosing a business intelligence tool, Tableau is the suitable business intelligence tool recommended. This is in addition to the fact

that Tableau can handle large volumes of data and is best at creating interactive visualizations (Lavanya, A., Sindhuja, S., Gaurav, L., & Ali, W. 2023).

Suggested Data Warehouse Architecture for Meier Drogeriemarkt GmbH

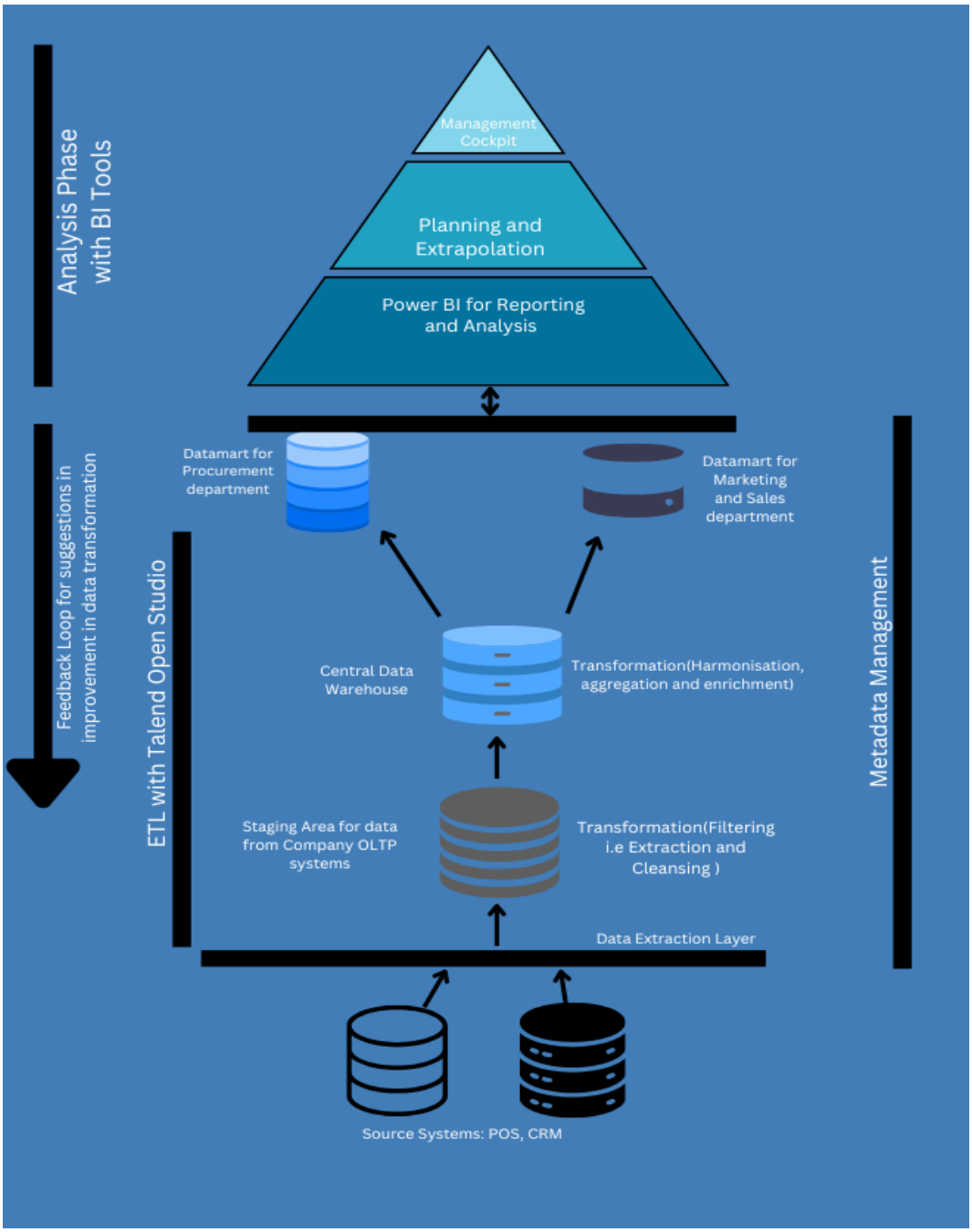

Fig._ 2: Referenced architecture (Dr. Vaas, Silke .Business Intelligence I[Course Book].IU International University of Applied Science. https://www.iu.org/ 2024).

ETL Process at Meier Drogeriemarkt GmbH Using Talend

Besides the description of Talend already stated above, this platform can be used for data integration, migration, building ETL processes, and Data warehousing.

Considering the data sources and volume of data at Meier Drogeriemarkt GmbH, Talend will be used to load relational database tables into the data warehouse, following the ROLAP concept.

Extraction

In the extraction phase data from Drogeriemarkt's source Systems e.g. CRM, databases, are extracted and ingested in the database of the staging area. In this process, the data from the source systems are not altered to ensure data consistency between the source systems and the staging area. The staging area does not persist data and facilitates rollback from the staging area to the source system in case errors in the data are detected that need to be corrected in the source system (Sreemathy, J., Nisha, S., & RM, G. P., 2020, March). Given the volume of data that will be handled, extraction with update notifications will be implemented. Once the data gets to the staging area, the transformation process will begin.

Transformation

This is the most complex and time-consuming part of the data, even though it largely depends on the nature of the data in the staging area. During transformation, different operations can be done on the data including cleansing, filtering, data standardization, data flow validation, data threshold validation, row and column transposing, joining, splitting, and sorting (Sreemathy, J., Nisha, S., & RM, G. P., 2020, March). Typically, the transformation process is composed of four components namely: Filtering, Harmonisation, Aggregation, and Enrichment.

- In the Filtering process, data is extracted from the staging area and cleansed. In cleansing content and technical defects are detected and corrected.
- In Harmonisation the cleansed data is integrated into a unified format. For dates written in the same format, Genders are only represented by M, W, and D for Male, Woman, and Diverse respectively. If necessary surrogate keys are created at this stage.
- During data aggregation is reduced to an agreed granularity or hierarchical level. Aggregation is related to defining dimensions and calculations. The calculations are based on key business figures. In the case of Drogeriemarkt, some key figures include: sales and payback points.
- During enrichment, the key figures specified in the aggregation are calculated to determine facts that can be analyzed based on the specified dimensions.

Loading

This is the last stage of the ETL Process and handles the part where data is written into the central data warehouse. Using Talend's graphical user interface, the ETL Process can be scheduled and monitored to ensure that the process runs smoothly.

Modelling a Datamart at Meier Drogeriemarkt GmbH

One purpose of building a data warehouse is to enhance the analysis of data it stores. To achieve this, data marts can be introduced to enable different departments at Meier Drogeriemarkt to have access to data only related to their business activities. The data mart will be modelled inform of a star schema where each schema will only contain facts and dimensions related to the department for which it is built. A star schema is an architecture used in data warehouses to model multidimensional spaces. Through this, branch managers will benefit from domain-specific analysis of their data which leads to fast and independent decision-making. In Fig._3, the data mart of the Sales and Marketing department at Meier Drogeriemarkt is presented.

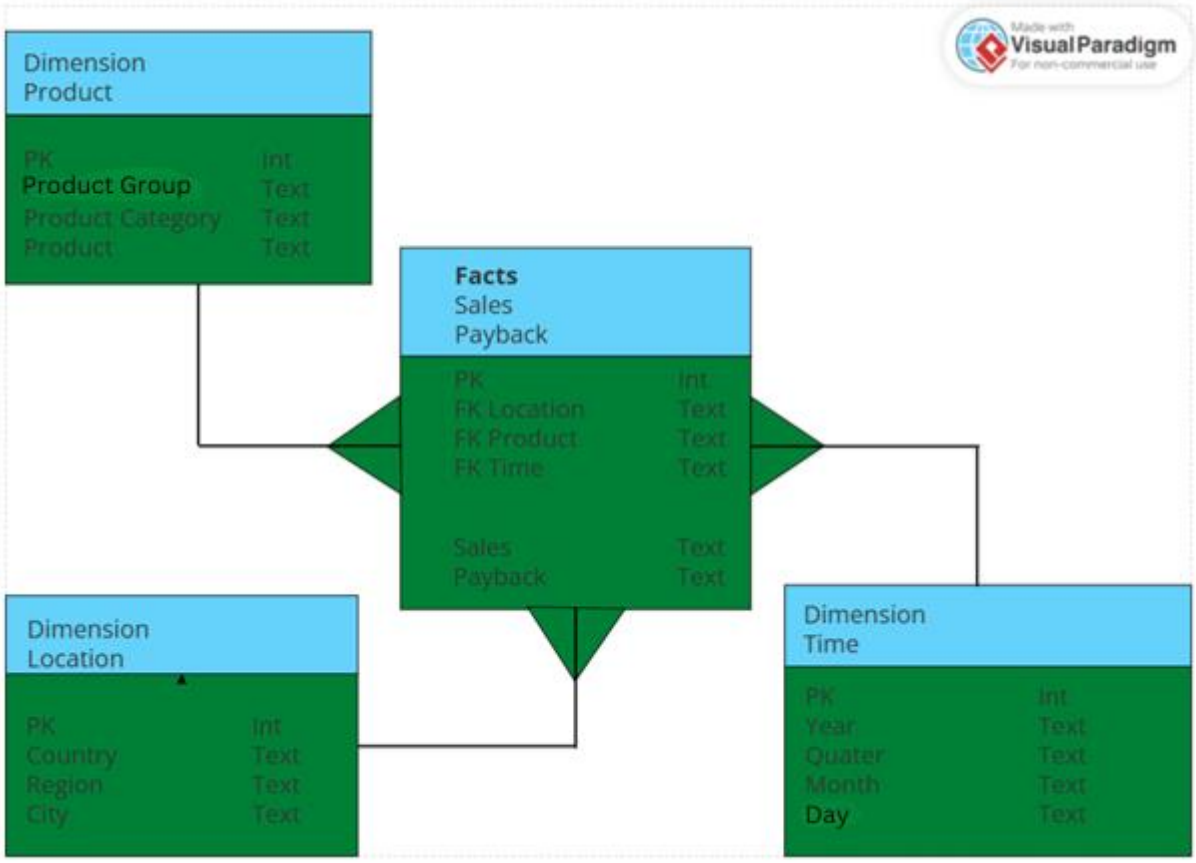

Fig._ 3: Star Schema Architecture for Sales and Marketing Departmental Data Mart at Meier Drogeriemarkt GmbH.

Reporting with Tableau

A reporting system aims to perform analysis of data to gain insights that help business managers make informed decisions towards the achievement of business objectives. In the context of Meier Drogeriemarkt GmbH, Tableau is chosen as the best tool for analyzing the data stored in the star schema of the data warehouse. Meier Drogeriemarkt will get the required licenses from Tableau that will enable them to use the Tableau cloud. Once this is done, Users in this case Analysts, will use the tableau connector that will enable them to have access to the required data in the data warehouse. Depending on the needs of the business branch managers, the analysts will create Dashboards or balanced scorecards around the key performance indicators of Meier Drogeriemarkt. For instance, the branch managers may be interested to

monitor the sales figures of a particular product category over a given period to analyse trends or make comparisons over days, months or years. Understanding trends can form the basis for developing marketing strategies that are customer centric to enhance the relationship between Meier Drogeriemarkt and its customers. These dashboards or scorecards will be shared with business managers after quality assurance is completed. Quality assurance is done to ensure that the information displayed on the dashboards or scorecards is accurate enough to support decision-making. The Analysts will support the business managers to understand and interpret the displayed data in the form of visuals, tables, or matrices. On the other hand, it is the role of the business managers to make decisions as they know the business requirements and targets that need to be met.

Challenges

The criteria for the comparison of ETL and Business Intelligence tools to determine the best tools to fulfill the business requirements of Meier Drogeriemarkt GmbH were completely based on a literature review. This alone might not be sufficient due to technological advancements that developers of these tools might have made from the time the researches were done. This implies that some of the futures or functionalities of the tools that were mentioned in the literature might be retired or improved. For this reason, the comparison might not be based on the current state of the tools. Another important resource that was not mentioned by Meier Drogeriemarkt is the budget available for the development and implementation of the reporting system. The absence of this constraint poses a challenge in the design of the entire reporting system because there is a risk of designing a system that might go beyond the amount Meier Drogeriemarkt GmbH is willing to invest or a system that might fall short of the expectations of this company. However, in designing this system, budget was not considered as a factor that would prevent the implementation of this reporting system.

Conclusion

In this report, a data warehouse architecture for Meier Drogeriemarkt was designed and the ETL process as well as the reporting approach was described. This was done based on the sets of tools that were chosen considering the comparison criteria from literature reviews and the business requirements that were outlined by Meier Drogeriemarkt GmbH. By adopting this reporting system at Meier Drogeriemarkt GmbH, the branch managers will be able to have a profound understanding of the business activities by analysing data from different sources. Hence, making decisions for the development of the business out of facts instead of intuition.

REFERENCES

Biplob, M. B., Sheraji, G. A., & Khan, S. I. (2018, October). Comparison of different extraction transformation and loading tools for data warehousing. In 2018 international conference on innovations in science, engineering and technology (ICISET) (pp. 262-267). IEEE.

Deng, S., Moutinho, L., & Meidan, A. (1991). Bank Branch Managers in a Marketing Era: Their Roles andFunctions in a Marketing Era. International Journal of Bank Marketing, 9(3), 32-38.

Dussart, C. (1998). Category management:: Strengths, limits and developments. European Management Journal , 16 (1), 50-62.

Hanine, M., Boutkhoum, O., Tikniouine, A., & Agouti, T. (2016). Application of an integrated multi-criteria decision making AHP-TOPSIS methodology for ETL software selection. SpringerPlus, 5, 1-17.

Hassan, R. S., Nawaz, A., Lashari, M. N., & Zafar, F. (2015). Effect of customer relationship management on customer satisfaction. Procedia economics and finance, 23, 563-567.

Lavanya, A., Sindhuja, S., Gaurav, L., & Ali, W. (2023). A comprehensive review of data visualization tools: features, strengths, and weaknesses. Int. J. Comput. Eng. Res. Trends, 10(01), 10-20.

Majchrzak, T. A., Jansen, T., & Kuchen, H. (2011, March). Efficiency evaluation of open source ETL tools. In Proceedings of the 2011 ACM symposium on applied computing (pp. 287-294).

Skender, F., & Manevska, V. (2022). Data Visualization Tools-Preview and Comparison. Journal of Emerging Computer Technologies, 2(1), 30-35.

Sreemathy, J., Nisha, S., & RM, G. P. (2020, March). Data integration in ETL using TALEND. In 2020 6th international conference on advanced computing and communication systems (ICACCS) (pp. 1444-1448). IEEE.

Szilveszter, M., Molnár, G., & Nagy, E. (2024, April). Comparison of commercial and open source ETL tools. In 2024 IEEE 11th International Conference on Computational Cybernetics and Cyber-Medical Systems (ICCC) (pp. 000037-000042). IEEE.

Thomsen, C., & Pedersen, T. B. (2005). A survey of open source tools for business intelligence. In Data Warehousing and Knowledge Discovery: 7th International Conference, DaWaK 2005, Copenhagen, Denmark, August 22-26, 2005. Proceedings 7 (pp. 74-84). Springer Berlin Heidelberg.